

JEE MAIN REVISION MASTER: POLYMERS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 79 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Polymers — Core Principles & Governing Laws	70 PYQs	Core Principle
High	Polymers — Applied Cases, Reactions & Traps	8 PYQs	Applied Topics
Standard	Polymers — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Polymers — Core Principles & Governing Laws	$\text{TiCl}_4, \quad n = \frac{963}{107} = 9, \quad \text{Al}(\text{Et})_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Polymers — Core Principles & Governing Laws (Very High Yield — 70 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Polymers.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Polymers.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Polymers PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C)..." [Ans: A]
- [2023 Morning Shift] Q#1 (MCQ): "Polymer used in orlon is :..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "Match List I with List II LIST I LIST II A. Weak intermolecular forces of attraction I. Hexamethylenediamine + adipic acid B. Hydrogen bondi..." [Ans: N/A]

3.2 Polymers — Applied Cases, Reactions & Traps (High Yield — 8 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Polymers.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2023 Evening Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : Ethene at 333 to 343K and 6 - 7 atm pressure in the presence of $AlEt_3$ and $TiCl_4$..." **[Ans: N/A]**
- **[2023 Morning Shift] Q#4 (MCQ):** "Match List I with List II List I Polymer List II Type/Class (A) Nylon-2-Nylon-6 (I) Thermosetting polymer (B) Buna-N (II) Biodegradable poly..." **[Ans: N/A]**
- **[2022 Morning Shift] Q#2 (MCQ):** "Which of the following is an example of polyester?..." **[Ans: N/A]**

3.3 Polymers — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Polymers.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2023 Morning Shift] Q#2 (MCQ):** "The polymer X - consists of linear molecules and is closely packed. It is prepared in the presence of triethylaluminium and titanium tetra..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C) Caprolactam (III) Teflon (D) Isoprene (IV) Nylon-6 Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) **[Correct]**
- (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Correct Answer: A

Solution Analysis: The correct match between List I (Monomer) and List II (Polymer) is as follows: (A) Tetrafluoroethene → (III) Teflon (B) Acrylonitrile → (I) Orlon (C) Caprolactam → (IV) Nylon-6 (D) Isoprene → (II) Natural rubber So, the correct answer is: A-III, B-I, C-IV, D-II

Benchmark Problem 2: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

Given below are two statements : Statement I : Ethene at 333 to 343K and 6 - 7 atm pressure in the presence of AlEt_3 and TiCl_4 undergoes addition polymerization to give LDP. Statement II : Caprolactam at 533 - 543K in H_2O through step growth polymerizes to give Nylon 6. In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and Statement II are true
- (B) Statement I is true but Statement II is false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are false

Correct Answer: None

Solution Analysis: Statement I is false. Ethene (ethylene) under certain conditions (high pressure, high temperature) in the presence of catalysts does undergo addition polymerization, but it forms polyethylene, not LDP (Low-Density Polyethylene) specifically. The formation of low-density polyethylene (LDPE) vs high-density polyethylene (HDPE) depends on the specific conditions and catalysts used. The catalysts mentioned (AlEt_3 and TiCl_4) are associated with the production of HDPE through the Ziegler-Natta process, not LDPE. Statement II is true. Caprolactam polymerizes to form Nylon 6 via a ring-opening poly

Benchmark Problem 3: JEE Main 2023 (Online) 11th April Morning Shift (Q#2)

The polymer X - consists of linear molecules and is closely packed. It is prepared in the presence of triethylaluminium and titanium tetrachloride under low pressure. The polymer X is-

- (A) High density polythene
- (B) Polytetrafluoroethane
- (C) Low density polythene
- (D) Polyacrylonitrile

Correct Answer: None

Solution Analysis: The polymer X described in the question is synthesized using triethylaluminium and titanium tetrachloride under low pressure. This is a clear indication that the Ziegler-Natta catalyst is used, which is typically employed in the synthesis of high-density polyethylene (HDPE). High-density polyethylene is made up of linear molecules that pack closely together, giving the material its high density. This is in contrast to low-density polyethylene (LDPE), which has a branched structure that prevents the chains from packing closely together. So, the polymer X is : Option A: High-density polyethylene